Andre Nogueira de Souza

Scientific Machine Learning & Computational Physics

Technical leadership | Research to production | GPU / HPC

Greater Boston, MA | andrenogueirasouza@gmail.com

Website | LinkedIn | Google Scholar

Profile

Scientific machine learning leader with a PhD in applied mathematics and 10+ years of experience across industry and academia. Combines mathematical depth, computational physics, and software engineering to solve challenging modeling problems. Experience spans industrial AI at Rescale and interdisciplinary research at MIT, with strengths in method development, research leadership, and translating scientific results into practical engineering capabilities.

Professional Experience

Senior HPC Engineer, Data and AI 2025–Present
Rescale, Inc. *San Francisco / Remote*

- **Research to product:** Developed and productized transformer-based surrogate models for engineering simulation. Outperformed relevant competing baselines on engineering datasets, helping win and expand engagements with leading global automotive manufacturers.
- **Technical direction:** Set research priorities and translate customer problems into technical plans. Shape product roadmaps and model-selection criteria with engineering, product, and commercial teams; guide programs through customer validation and production handoff.
- **Trustworthy evaluation:** Design checks for physical validity, uncertainty, distribution shift, and numerical artifacts so engineering decisions reflect model reliability and limitations.
- **Reusable workflows:** Productized training, evaluation, visualization, and inference tools with clear interfaces and defaults. Teach and mentor field and engineering teams to extend these workflows independently.
- **Model development:** Improved difficult engineering predictions where standard approaches were insufficient. Combined physics-aware features, residual modeling, targeted sampling, and loss-function design to turn underperforming models into useful customer-facing predictions.

Research Scientist 2019–2025
Massachusetts Institute of Technology *Cambridge, MA*

- **Research leadership:** Led interdisciplinary work across scientific ML, fluid dynamics, climate physics, stochastic systems, and uncertainty quantification. Led a 17-author, multi-institution program from formulation and software integration through validation and publication.
- **Scientific methods:** Developed generative and probabilistic models for physical systems, calibration methods for uncertain simulations, and reduced models of turbulent dynamics.
- **Research software:** Developed GPU algorithms for iterative linear solvers and discontinuous Galerkin methods. Contributed numerical formulations and research software for atmosphere and ocean simulation.
- **Mentorship and publication:** Supervised undergraduate, graduate, and postdoctoral researchers. Carried results through numerical verification and peer review, with publications in PNAS, Physical Review Letters, Journal of Fluid Mechanics, and other scientific journals.

Education

PhD, Applied and Interdisciplinary Mathematics 2016
University of Michigan

Technical Expertise

- **Scientific ML:** Transformers, neural operators, surrogate models, diffusion and score models, probabilistic forecasting, uncertainty quantification, and reduced-order modeling.
- **Evaluation:** Physical validity, benchmark design, model selection, distribution shift, sensitivity analysis, failure modes, reproducibility, and engineering-relevant metrics.
- **Physics and numerics:** Partial differential equations, fluid mechanics, dynamical systems, stochastic processes, numerical methods, and simulation integration.
- **Software and HPC:** Python, PyTorch, JAX, Julia, Rust, KernelAbstractions.jl, distributed training, GPU performance optimization, and scientific visualization.

Selected Publications

Complete list: Google Scholar publication profile.

- S. Bouabid, **A. N. Souza**, and R. Ferrari. Score-Based Generative Emulation of Impact-Relevant Earth System Model Outputs. Journal of Advances in Modeling Earth Systems, 18(3), e2025MS005558, 2026.
- L. T. Giorgini, F. Falasca, and **A. N. Souza**. Predicting Forced Responses of Probability Distributions via the Fluctuation-Dissipation Theorem and Generative Modeling. Proceedings of the National Academy of Sciences, 122(41), e2509578122, 2025.
- S. Silvestri et al. (including **A. Souza**). A GPU-Based Ocean Dynamical Core for Routine Mesoscale-Resolving Climate Simulations. Journal of Advances in Modeling Earth Systems, 17(4), e2024MS004465, 2025.
- L. T. Giorgini, K. Deck, T. Bischoff, and **A. Souza**. Response Theory via Generative Score Modeling. Physical Review Letters, 133, 267302, 2024.
- **A. Souza**. Representing Turbulent Statistics with Partitions of State Space I: Theory and Methodology. Journal of Fluid Mechanics, 997, A1, 2024.
- **A. N. Souza**, T. L. Lutz, and G. R. Flierl. Statistical Nonlocality of Dynamically Coherent Structures. Journal of Fluid Mechanics, 966, A44, 2023.
- **A. N. Souza** et al. The Flux-Differencing Discontinuous Galerkin Method Applied to an Idealized Fully Compressible Nonhydrostatic Dry Atmosphere. Journal of Advances in Modeling Earth Systems, 15(4), e2022MS003527, 2023.
- **A. N. Souza**, I. Tobasco, and C. R. Doering. Wall-to-Wall Optimal Transport: Theory and 2D Computations. Journal of Fluid Mechanics, 889, A34, 2020.

Selected Talks, Recognition & Service

- **Invited talks:** MIT PAOC Colloquium, MIT Ocean Engineering Seminar, MIT CSE Community Seminar, Nordita WINQ plenary, Brown LCDS, NYU Atmosphere Ocean Science Colloquium, and University of Michigan AIM Seminar.
- **Recognition and research support:** MIT-Google Program for Computing Innovation, NCAR CISL large allocation, Bringing Computation to the Climate Challenge, CliMA Research Scientist, and Woods Hole GFD Fellowship.
- **Service:** Journal peer review, curriculum development, technical interviews, open-source scientific software, and technical documentation across teams.